By Team Bronze - Old Dominion University - CS 410 - Spring 2026

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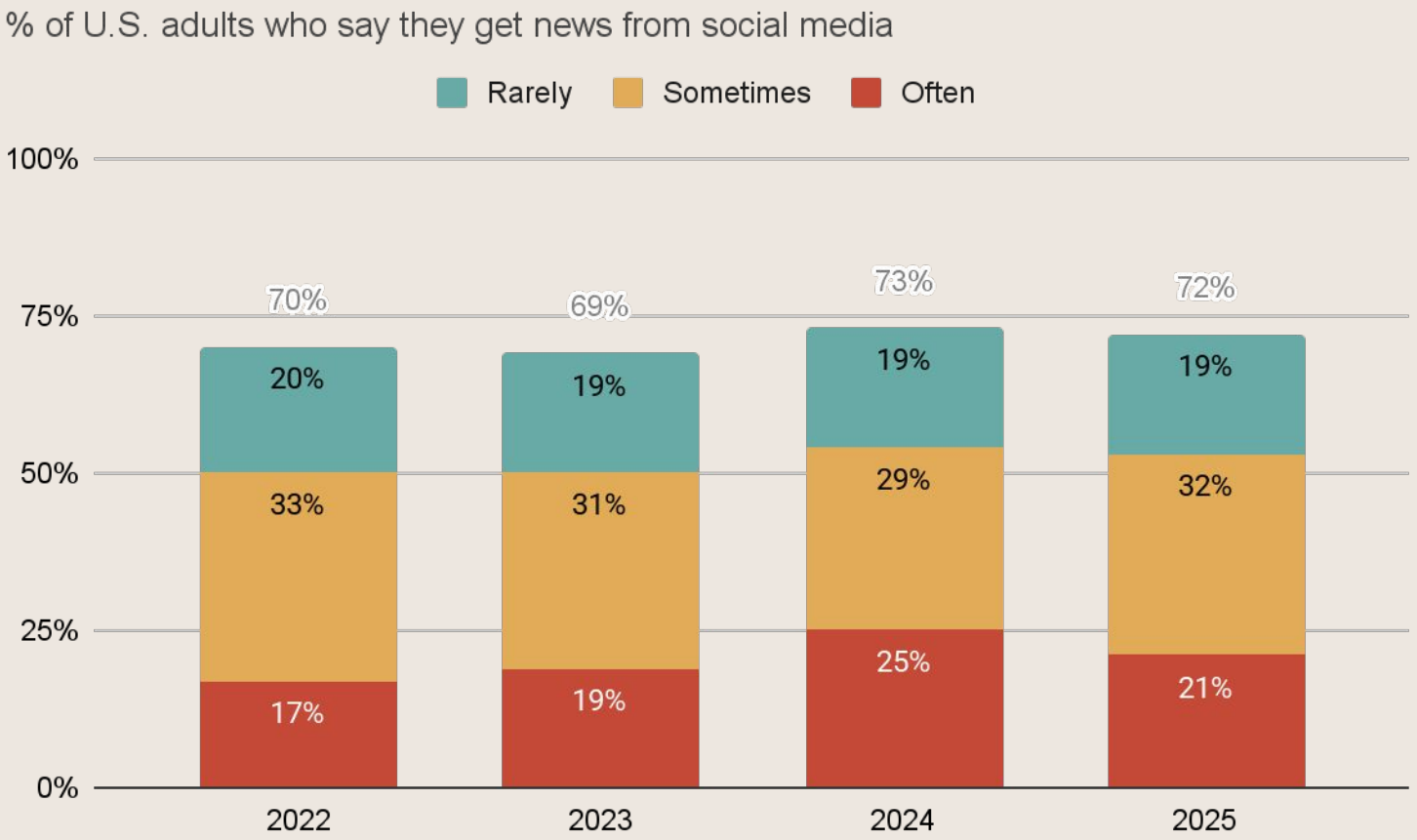
THE PROBLEM

Mis- and Disinformation are a rapidly growing issue in online spaces, with incorrect and highly biased information constantly being spread across social media platforms and other internet sites

Social Media as News

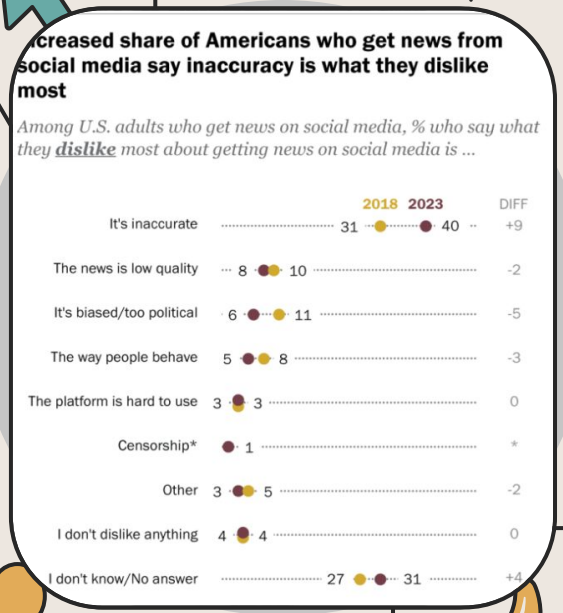


Data from Pew Research Center (2025, September 25)

Misinformation in our Digital Ecosystem

94% of people report encountering misinformation online (The Alan Turing Institute, 2024)

38.2% of U.S. news consumers report unknowingly sharing fake news. (Redline Digital, 2024)



DISTRUST IS GROWING

- Concerns about inaccurate information on social media are rising (Pew Research Center, 2024)
- 40% of Americans who get news from social media cite inaccuracy as their biggest concern. (Pew Research Center, 2024)
- Only 37% say they trust information from social media platforms. (Pew Research Center, 2023)

@REDFLAG

So why does it persist?

People don’t pay attention to what they’re consuming/sharing (Ordway, D 2021)

Social media platforms reward engagement over accuracy (Ceylan, G 2023)

Evidence shows it is primarily a mindless action, done out of passivity and without care, and current systems only encourage it

- Community based fact-checking is too slow, corrections don’t come in until long after posts have taken off (Blanding, M 2025)
- Professional human fact-checking is accurate but costly, and limited coverage makes real-time intervention impractical
- However, purely AI based fact checking often only compounds the problem as AI checkers have been found to be wrong 37-94% of the time (Ford 2025)
- Overconfident AI judgements can reinforce misinformation rather than correcting it
- Going to a separate website is time consuming and requires extra effort

Current fact-checkers aren’t effective

Why does this matter?

When misinformation spreads:

- Public trust declines (Pew Research Center, 2023)
- False information spreads unintentionally (Redline Digital, 2024), which causes countless negative outcomes including
  - Decline in public health (Denniss 2025)
  - Reduced trust in democratic institutions and increased polarization (Royal Holloway University of London 2024)
  - Hindering environmental efforts
  - Influencing the economy (Chen 2023)
  - And more

The fact checking tools of today are either too niche or bothersome to tackle this misinformation epidemic.

Problem Characteristics (THE GAP)

LACK OF UNIVERSAL TOOL

Many existing verification tools are platform-specific and not unified. For example: Community notes on X, NewsGuard for news sites, LLMs when manually provided context.

NO MOTIVATION TO VERIFY

Platforms and algorithms encourage instant sharing with one click, while verifying claims may take minutes. Rationally, users choose not to spend this time and effort.

LACK OF CREDIBILITY AWARENESS

Users receive no real-time credibility signals while browsing.

INCONVENIENCE AND DISCONNECTED VERIFICATION

Fact-checking often requires manual searching outside the platform.

ALGORITHMS REINFORCE BIAS

User's algorithms feed them content they inherently agree with which makes misinformation easier to spread.

RedFlag is a cross-platform fact checking (RedFlag) and bias indicator (BaisStar) running behind the scenes to tackle misinformation. When users are consuming the news any inaccuracies will be marked instantly with credible sources upon request, to prevent the unknowing spread of misinformation in today's information age. Furthermore on politically divisive issues RedFlag will star the content and offer different viewpoints from the left, right, and center at the users choosing.

THE SOLUTION

Solution Characteristics (CLOSING THE GAP)

ONE UNIVERSAL TOOL

A general tool for uncovering misinformation, no matter where the user browses.

EASILY ACCESSIBLE, UNPROMPTED AND INSTANT VERIFICATION

Users are warned and informed before sharing potentially false claims, discouraging impulsive sharing.

AWARENESS OF CREDIBILITY

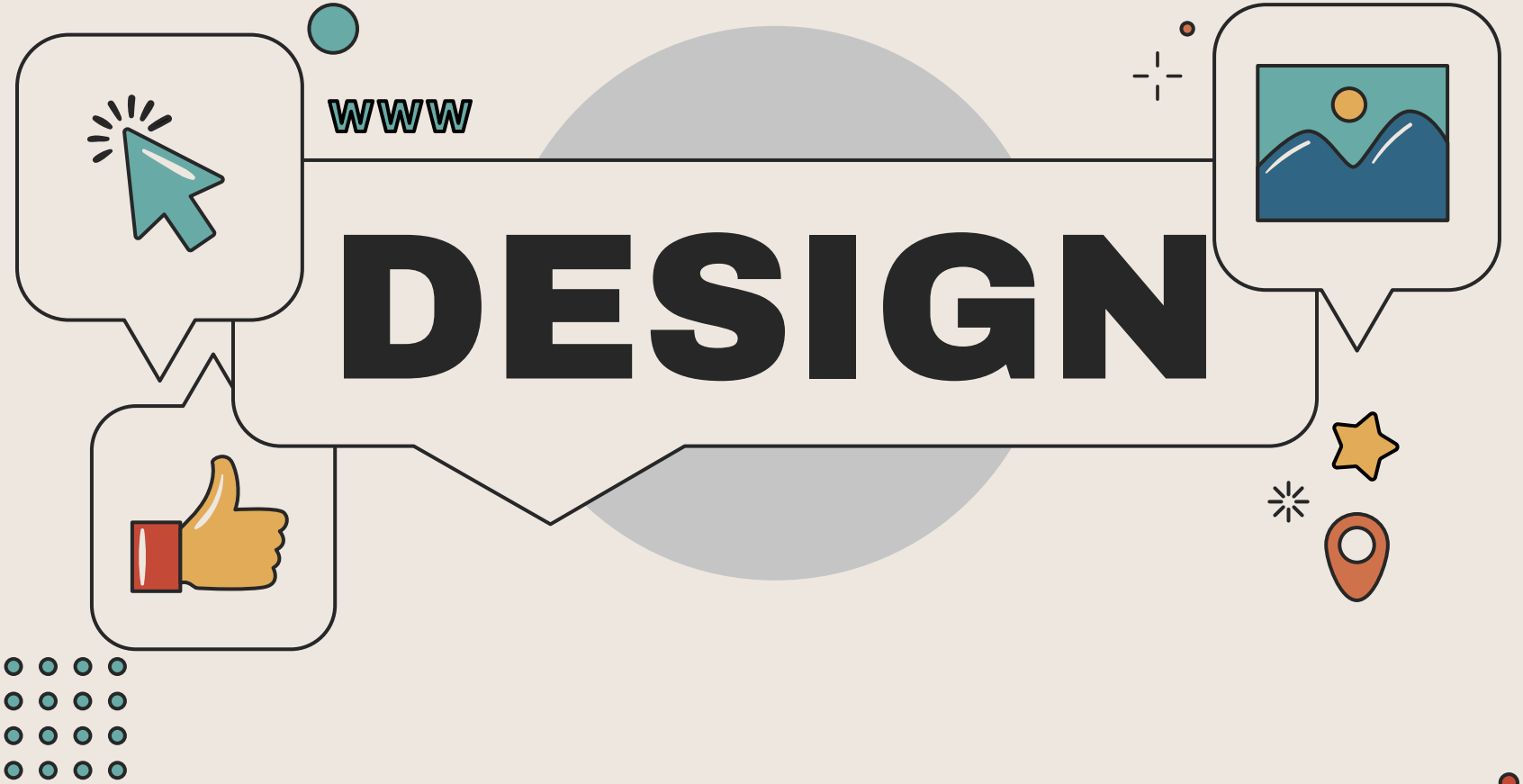
Users are actively informed of the credibility of sources they view while they're browsing.

CONVENIENCE AND VERIFICATION

Fact-checking is quick and painless, requiring little input from the user.

SHARING DIFFERENT POINTS OF VIEW

Users are provided with a multitude of sources leaning in different areas of the political spectrum.


www
DESIGN

HOW WILL REDFLAG WORK?

INTEGRATED DETECTION

Integrated by a chrome extension and or desktop application scanning the users screen to find misinformation at all times.

REAL-TIME DETECTION

Highlights questionable information immediately as it is presented on screen.

Also highlights political content to broaden users information scope and tackles social media algorithms.

FRICTIONLESS & COMPLETE

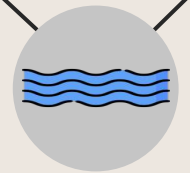
Highlights require no user input.

Can deliver detailed overview on request.

WHAT ARE REDFLAG'S DESIGN PRINCIPLES?

UNBIASED

- RedFlag will not support any specific political standpoint
- No explicit ideological labeling of content
- Evidence based evaluation
- On disputed topics, RedFlag will use multiple sources and perspectives



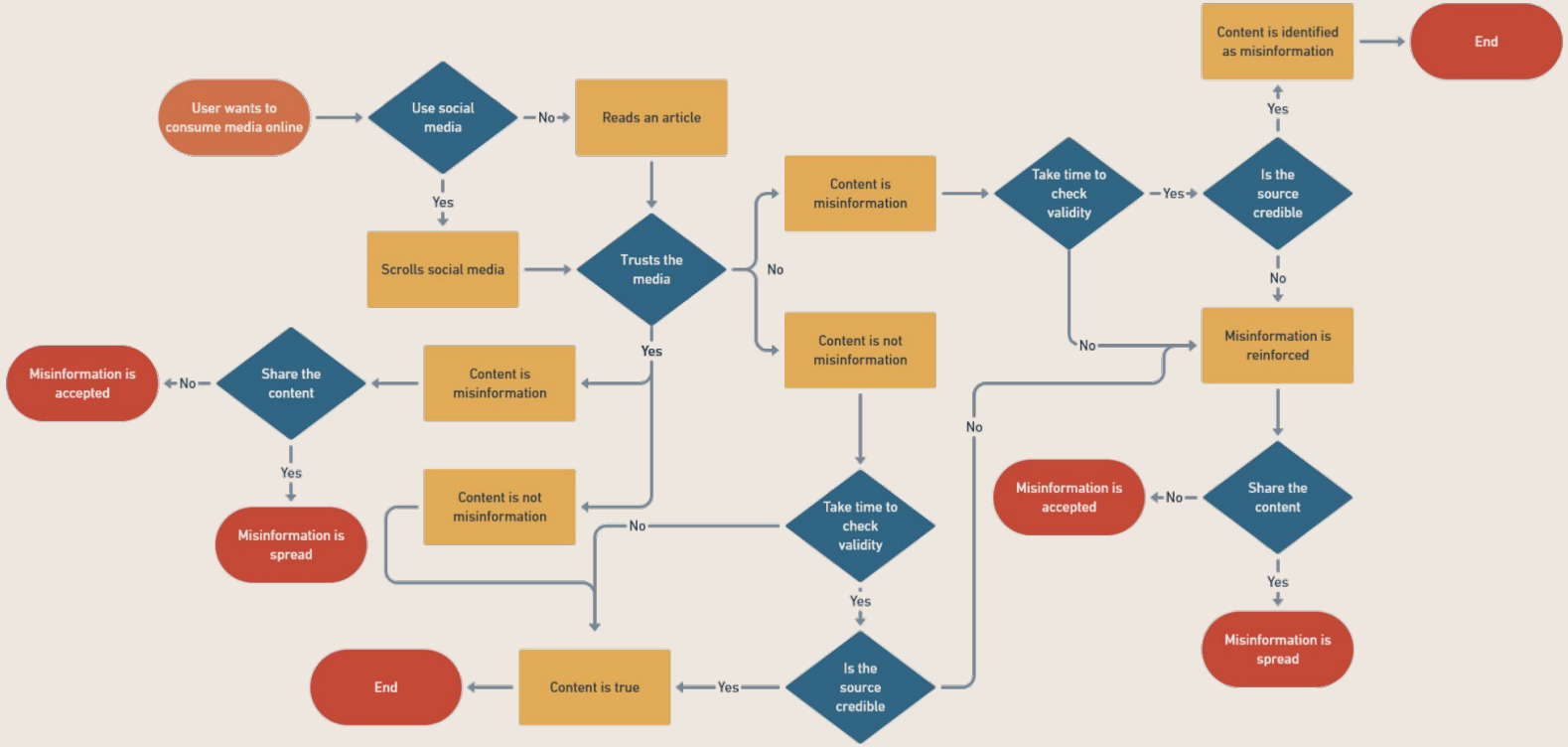
UNINTERRUPTED FLOW

- Pop-ups would interrupt user's browsing experience
- Highlighting text is significantly less intrusive

NOT PROFESSIONAL REPLACEMENT

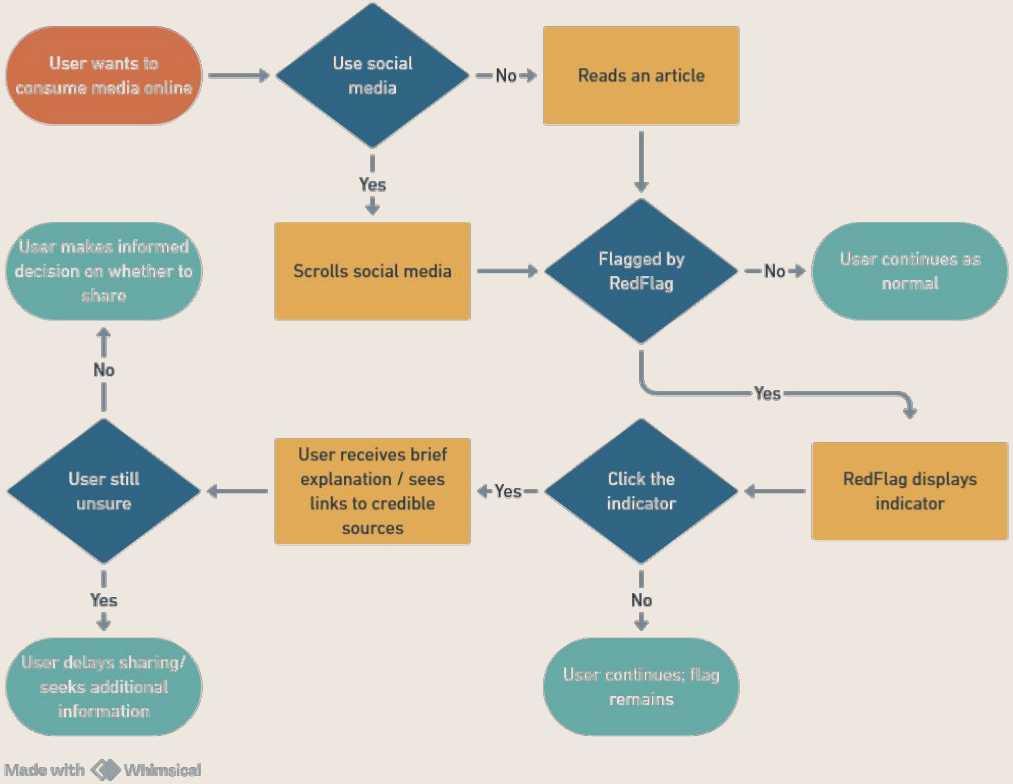
- Supports decision-making without dictating it
- Optional context is provided upon the user's request
- Supports journalism, but does not substitute it
- Does not act as a final authority on truth

Problem Process Flow

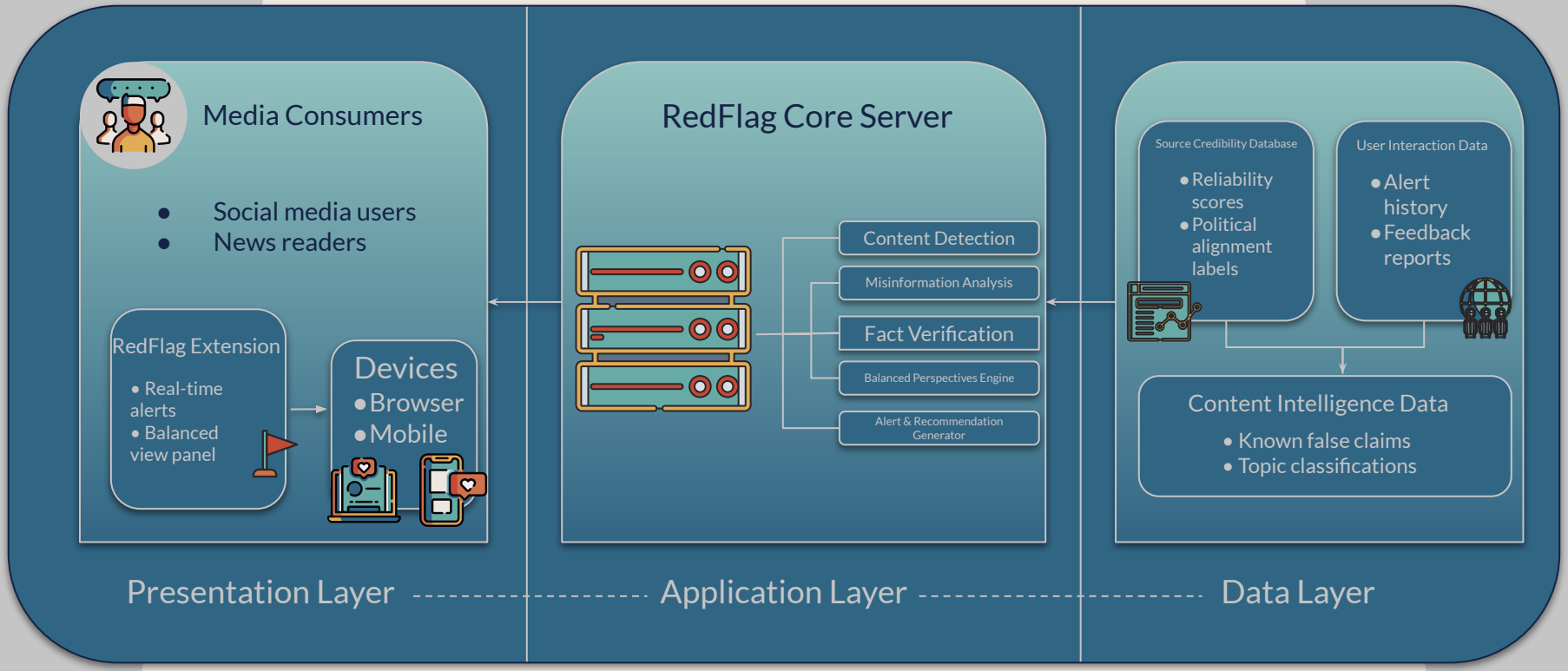


Made with Whimsical

Solution Process Flow



MAJOR FUNCTIONAL COMPONENT DIAGRAM



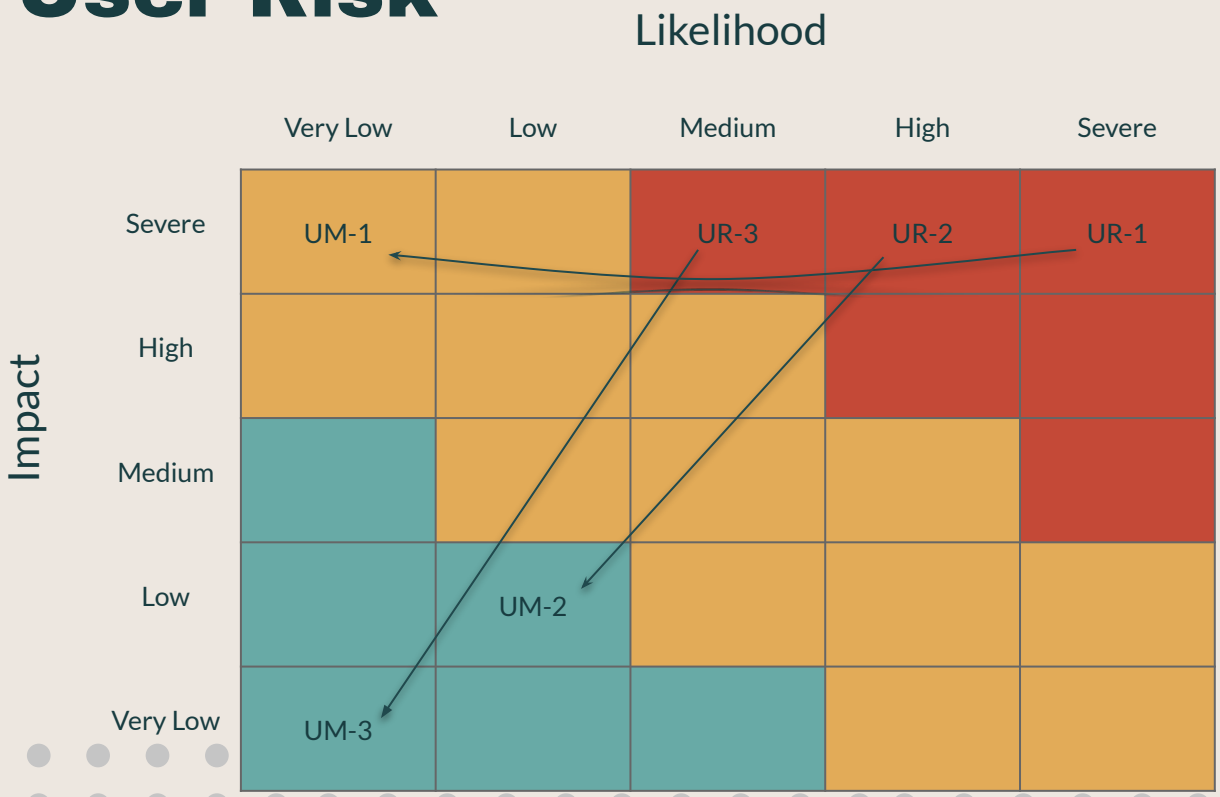
COMPETITORS TABLE

	Fact checker databases (Example: Google Fact Check Explorer)	News Guard	Community Notes	LLM
Features	-Search for stories by keyword, person, or image that have been debunked -Fact check other stories yourself	-Rates trustworthiness of a news source on a scale from 1-100 -Team of journalists fact checking	-Allows users to add contextual explanations or fact checks to posts -Other contributors rate notes for helpfulness, only helpful notes are shown	-Ask AI the veracity of a claim/source (e.g. @Grok is this true?) -AI uses its dataset to respond
Strengths	-Quickly able to search for rumors or claims being made online -Aggregate source of online disinformation	-Real human input on trustworthiness -Partnered with organizations for better fact checking	-Posts remain visible with extra context added -Uses a diverse group of users to reduce bias	-Ease of use- users already familiar with service, easily available, immediate response -Low manpower required
Weaknesses	-Allows biased sources to fact check other sources -Does not provide input on every claim made online -No verified source of claims	-Possible bias due to human fact checkers	-Slow response time makes notes appear long after a misleading post has spread -Organized groups can attempt to manipulate note ratings	-Frequently incorrect -Confidently wrong, fooling users -Dataset is fallible
RedFlag	Rather than search for stories, it will be real time when reading articles	Works more actively to inform the user rather than passively on standby to check	With the browser extension it will work for more than just online posts, and be faster due to a centralized database	Will use AI solely increasing work flow of fact checking, but not act as a source of authority on any subject matter

Legend

Low Risk | Medium Risk | High Risk

User Risk



Risks

UR-1: Misinformation is not flagged or correct information is flagged.

UR-2: Users question credibility and stop using the tool.

UR-3: Intrusive UI or slow performance leads to user abandonment.

Mitigations

UM-1: Continuous AI retraining with updated misinformation datasets and multiple fact-check sources.

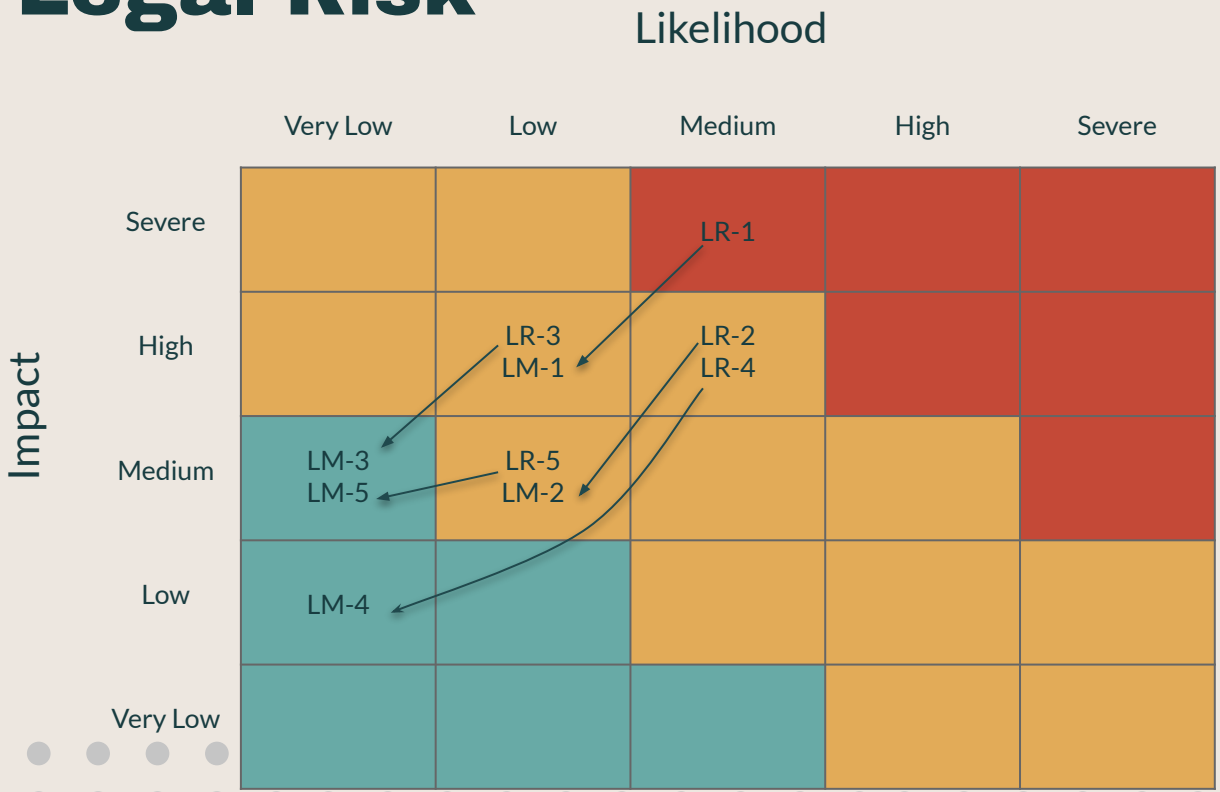
UM-2: “Why this was flagged” transparency feature with cited sources.

UM-3: Customizable alerts and performance optimization for smoother UX.

Legend

Low Risk	Medium Risk	High Risk

Legal Risk



Risks

LR-1: Improper handling of personal data
LR-2: Platform Terms of Service Violations resulting in removal from app stores
LR-3: Claims of reputational damage from incorrect flagging.
LR-4: Unauthorized use of protected datasets or APIs.
LR-5: Increased oversight of AI-based content moderation tools.

Mitigations

LM-1: Local data processing when possible and encrypted data transmission.
LM-2: Routine compliance reviews of browser store and API policies.
LM-3: Use of neutral phrasing such as “Potentially Misleading” instead of definitive claims.
LM-4: Use of licensed APIs and public domain datasets only.
LM-5: Clear disclaimer stating RedFlag is informational and not legal or professional judgment.LM-1: Local data processing when possible and encrypted data transmission.
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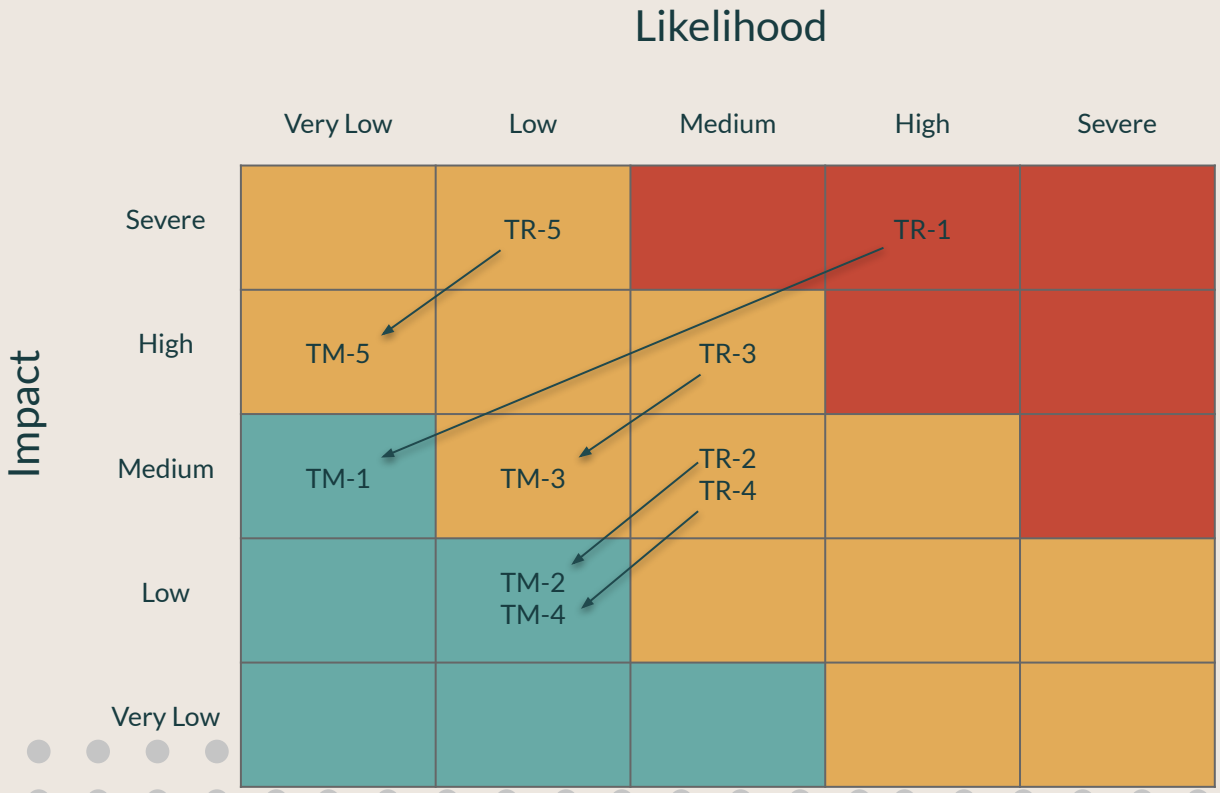
Legend

Low Risk	Medium Risk	High Risk

Technical Risk



Risks

TR-1: AI model produces biased or inconsistent results.

TR-2: API usage costs exceed projected operational budget.

TR-3: Slow processing or latency reduces user adoption.

TR-4: Extension malfunctions across different browsers or software updates.

TR-5: Security vulnerabilities expose the system to potential exploitation.

Mitigations

TM-1: Routine benchmarking, bias testing, and model performance audits.

TM-2: Rate limiting, caching frequent queries, and cost monitoring tools.

TM-3: Asynchronous API calls and backend optimization.

TM-4: Cross-browser testing and staged rollout deployment.

TM-5: Code audits, encryption protocols, and regular penetration testing.

User Roles

Media consumer - Consumes the misinformation while having RedFlag on mobile or desktop/laptop.

Admin - Quality control, source manual verification ext.

User Stories (Media Consumer)

- As a media consumer, I want RedFlag to automatically analyze content while I browse so that I don’t have to manually check for misinformation.
- As a media consumer, I want to see a warning when content may be misleading so that I can be cautious while reading.
- As a media consumer, I want a clear explanation of why content was flagged so that I understand the reasoning behind the warning.
- As a media consumer, I want to report incorrect flags so that the system becomes more reliable.
- As a media consumer, I want to choose when to expand or ignore warnings so that I stay in control of my experience.
- As a media consumer, I want to enable or disable RedFlag on certain websites so that I can customize where it runs.
- As a media consumer, I want RedFlags chat assist to walk me through the information I am seeing and to better understand how to avoid misinformation.

User Stories (Admin)

- As an admin, I want to review content that is flagged as inaccurate so that I can correct or validate the system's analysis.
- As an admin, I want to update or override misinformation labels so that incorrect results do not harm user trust.
- As an admin, I want to manage the list of trusted and comparison sources so that viewpoint recommendations remain balanced and credible.
- As an admin, I want to assign credibility scores and viewpoint labels to sources and individuals so that the system can recommend them accurately.
- As an admin, I want to monitor false positives and false negatives so that I can improve system accuracy.
- As an admin, I want to view system activity and flagged content statistics (bias and credibility score) so that I can understand how the platform is performing.

Feature Table

Category	Features	Media Consumer	Admin
Content Analysis	Highlight Claims	✔	✔
	View Sources	✔	✔
	Multi-Perspective / Balanced Perspectives Engine	✔	✔
	Source Credibility Rating	✔	✔
User Interaction	Click to expand verification details	✔	
	Toggle RedFlag on specific websites	✔	
Feedback System	Report Incorrect Flags	✔	✔
	View explanation for flagged content	✔	✔
	View confidence level	✔	✔

Feature Table

Category	Features	Media Consumer	Admin
Administration	Review flagged content and user feedback		✔
	Override or update misinformation labels		✔
	Manage trusted / comparison sources		✔
Monitoring	Track false positives/negatives		✔
	View system activity and performance stats		✔

User Interface Mockups

Browser extension or desktop application?
Mobile app? (questions from cameron)

Work Breakdown Structure

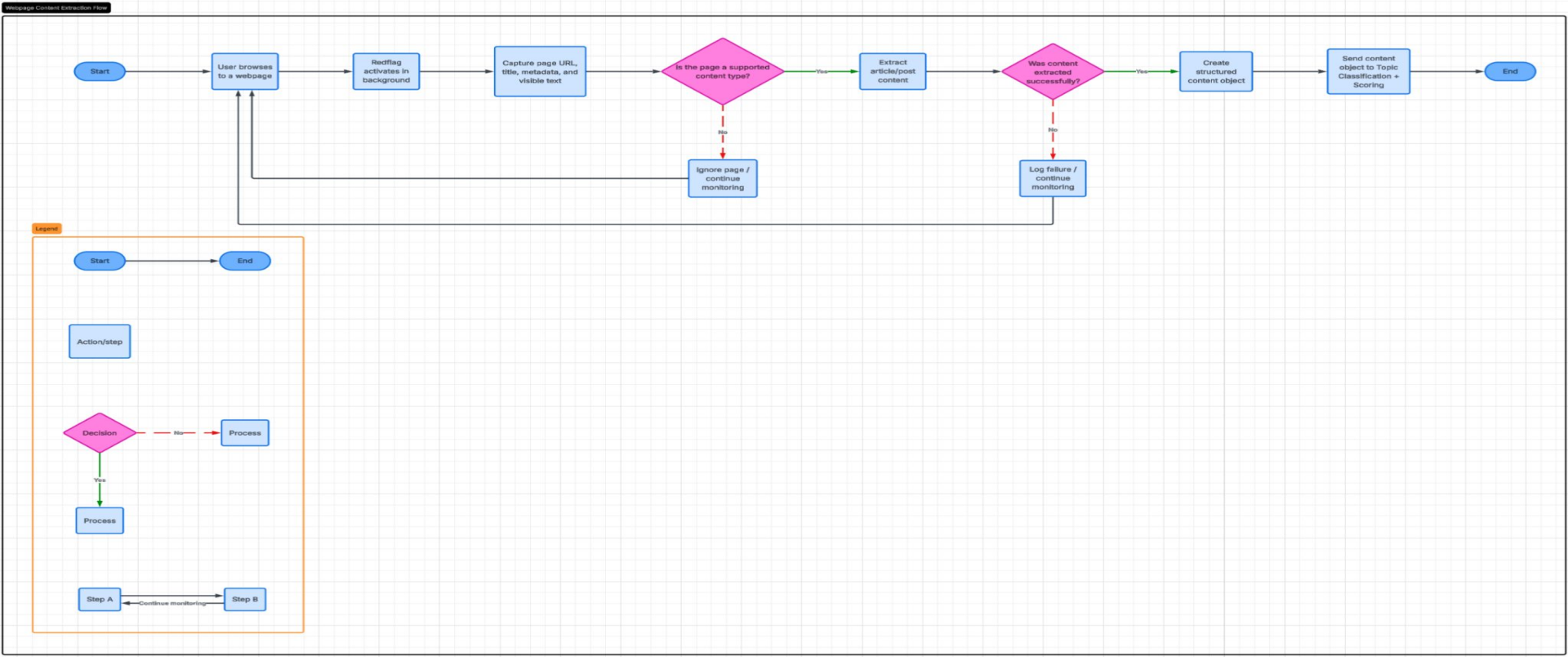
- Frontend
  - UI screens
  - Components

- Backend
  - Misinformation identification engine
  - Recommendation engine (bias star)

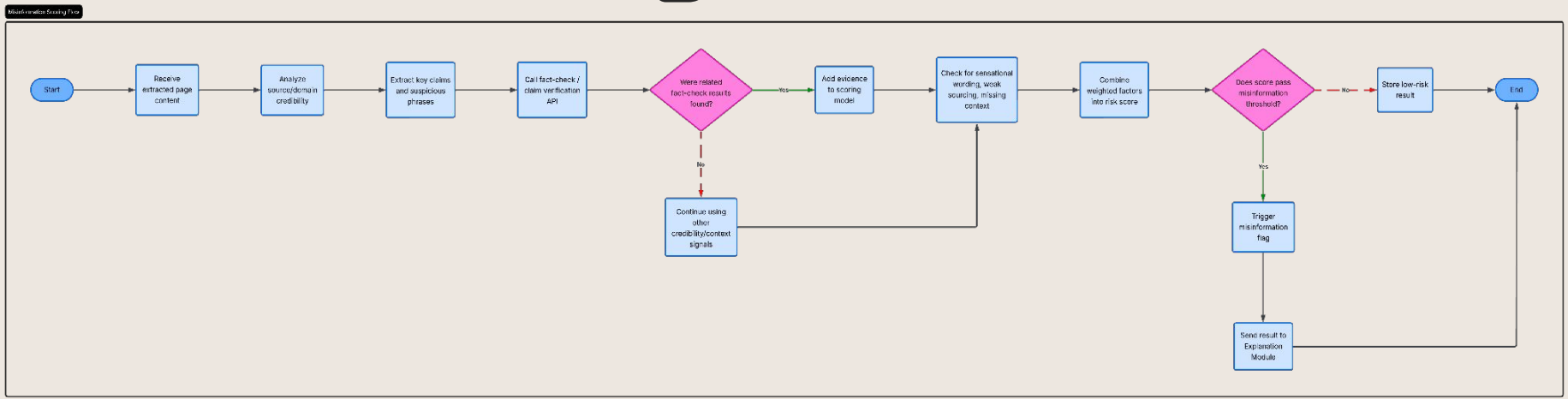
- Database
  - Tables
  - Relationships
  - User experience
- APIs
  - Fact checking API
  - News API

https://docs.google.com/document/d/166_ItRCBmN98U2SZCdYwReS59W4GzJxip8P4oEsoxZk/edit?tab=t.0#heading=h.25rap0dktljw

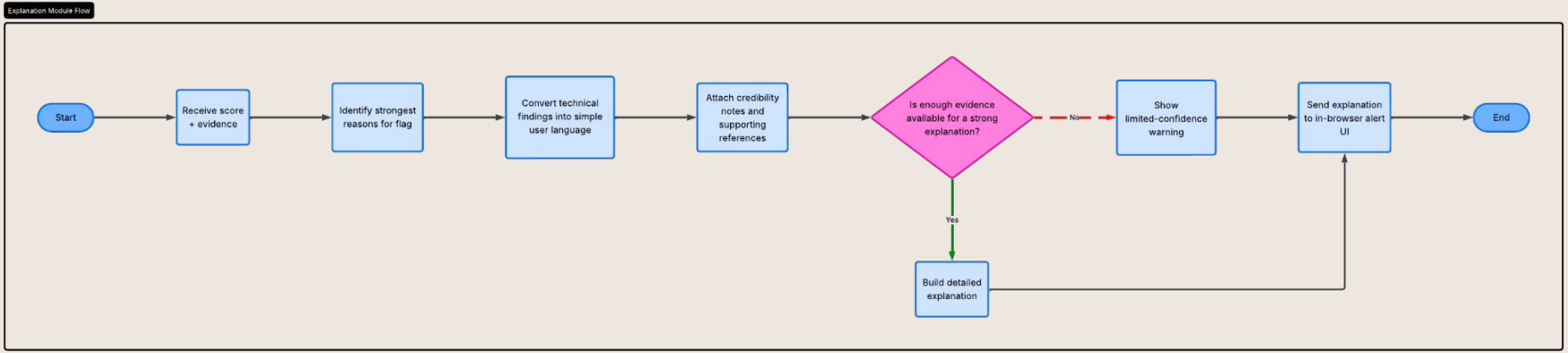
Content viewing Algorithm



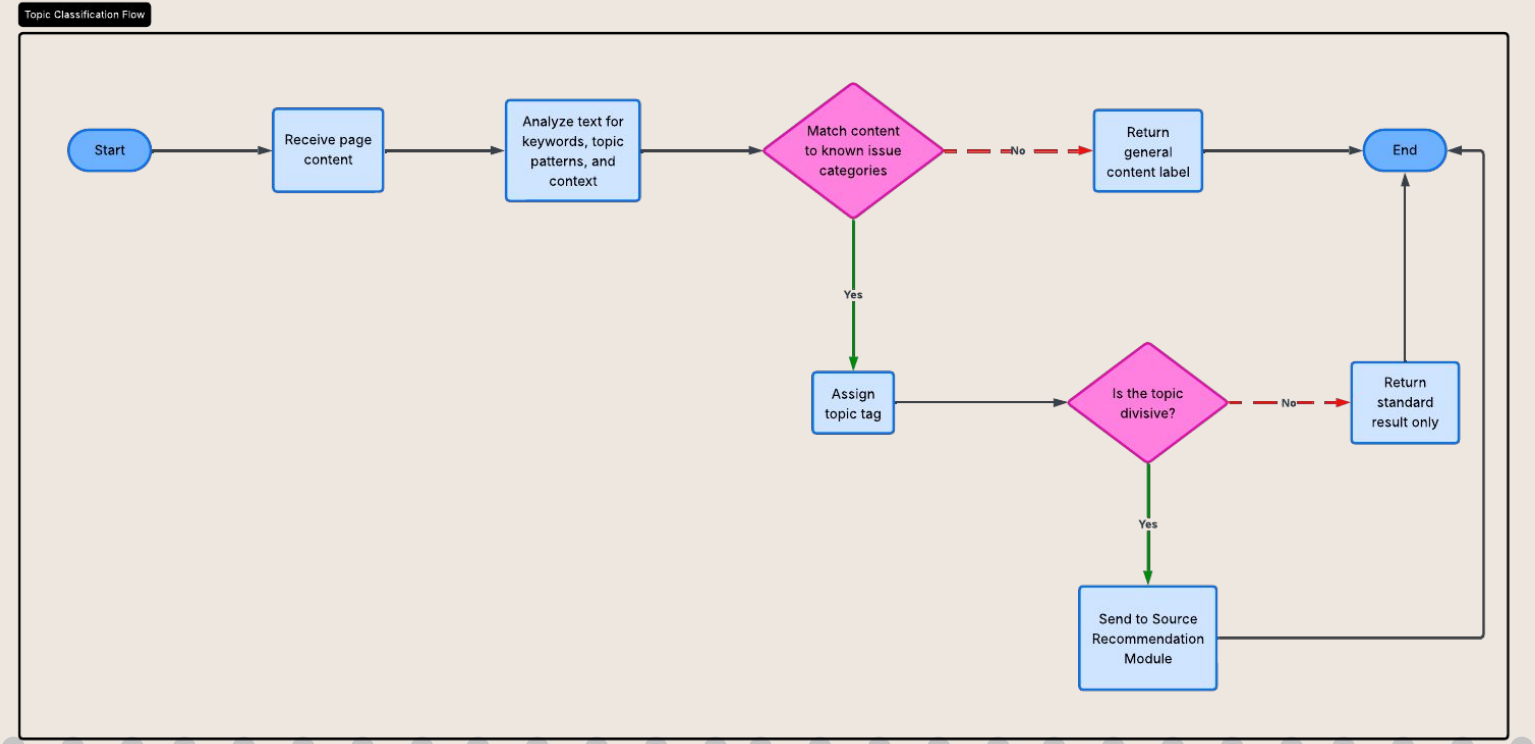
Misinformation Scoring algorithm



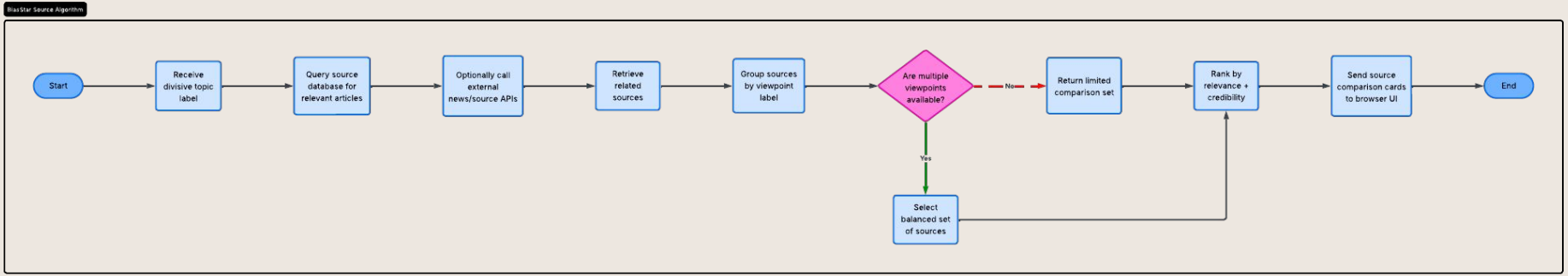
Explanation Algorithm

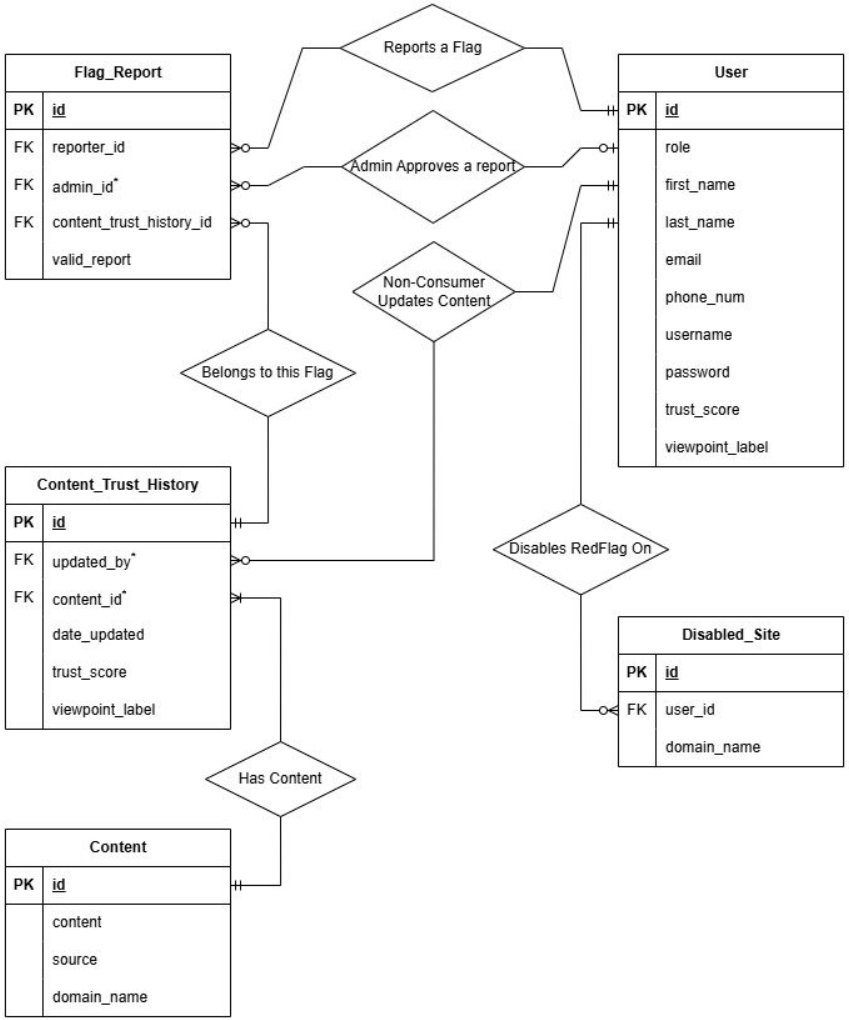


Topic Classification Algorithm



BiasStar Source Algorithm





Database Schema

- Redflag misinformation identification engine will be user 0 for the purpose of flagging content
- Extracted article text is stored in the content field of the Content table, and the URL source of the content is stored in the source field.
- Any time content is updated with a score or political alignment label by either the misinformation engine or an admin, a new record with that information and a timestamp will be recorded in the Content_Trust_History table.
- Content that is updated to be under a certain trust_score threshold will be considered “flagged.”
- Flag_Report contains an already flagged content that a user (admin or media consumer) has reported as inaccurate. Valid_report is a binary column that displays 1 if the report is a valid report (meaning inaccurate content), and 0 if an admin has altered the report and corrected it or verified it.

External Interfaces

APIs

- **Fact-checking**
  - **This will be used for taking in questionable information, and comparing against our database**
- **News**
  - **This will take in public news information and be categorized by topic**
- **LLM**
  - **This will take in all information in our database and parse it to summarize findings**
- **Interface**
  - **This will scan information from users' browsing and verified**

Development Tools & Dependencies

Development:

- Version Control
  - Git (Github)
- IDE
  - VSCode
- Builds
  - Gradle (Subject to change)

Dependencies:

- Front-end
  - Javascript for browser extensions
- Back-end
  - Python for handling information data sets
- Database
  - PosgreSQL

Sprint Breakdown

Sprint 0
Weeks 1-3

Planning & Setup

- Define system scope and requirements
- Set up development environment & repository
- Research fact-checking sources and APIs

Sprint 1
Weeks 4-6

Core Infrastructure

- Build UI prototype
- Implement basic text detection & highlighting
- Set up backend devices

Sprint 2
Weeks 7-10

Fact-Checking Integration

- Integrate external fact-checking APIs and sources
- Implement claim detection pipeline
- Display evidence and source links

Sprint 3
Weeks 11-13

User Interaction

- Click-to-expand verification
- User feedback system
- Handle uncertainty

Sprint 4
Weeks 14-16

Refinement & Testing

- Improve accuracy
- UI/UX improvements
- Performance testing and system optimization

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Glossary

- **Friction** - something that causes an application to not be seamless while working.
- **L.L.M.** - A Large Language Model (also known as a ChatBot) such as ChatGPT, Gemini, or Claude.
- **Misinformation** - Provably false or inaccurate information
- **Disinformation** - False or inaccurate information that is shared with the intention to mislead others
- **Confirmation Bias** - The tendency to interpret new information and evidence as confirmation of one's existing beliefs or theories, often while ignoring or discounting opposing evidence
- **Credibility Score** - A score given to a claim that is found online rating the credibility of the claim, or the likelihood that the claim is correct.

THANKS!

Do you have any questions?

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